

HOWTO — AIStudio User Guide

Practical answers for day-to-day AIStudio use. Not a getting-started guide (see [QUICKSTART.md](#)) — reach for this when you need to do something specific or something isn't working as expected.

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Shell & Terminal

What is the shell? The shell (Terminal) is a text-based interface for running commands on your Mac. AIStudio setup and management uses the shell for tasks the UI doesn't cover — like installing models or ingesting a new corpus. New to the shell? See [Apple's Terminal User Guide](#).

How do I open the Terminal? Press `Cmd+Space`, type `Terminal`, press Enter. Or open Finder → Applications → Utilities → Terminal.

Why does my terminal say "command not found" when I paste commands? Two common causes: - You pasted a `#` comment line — zsh tries to execute it. Paste only the commands, not the comment lines. - Your AIStudio aliases aren't loaded yet. Run:

```
source ~/.zshrc
```

Then try the command again on its own line. Do not chain: `source ~/.zshrc && ais_start` will fail. Run `source ~/.zshrc` first, then `ais_start`.

Using AIStudio

After running `./ais_install` from the repo root (see QUICKSTART.md), these commands are available from any terminal. If a command is not found, run `source ~/.zshrc` first.

Command	What it does
<code>ais_install [cmd]</code>	Install AIStudio user commands — <code>ais_install ais_log</code> adds a single command, <code>ais_install --verify</code> checks all aliases
<code>ais_start</code>	Start all services and open the UI in your browser
<code>ais_stop</code>	Stop all services
<code>ais_bench</code>	Run a benchmark on the demo corpus
<code>ais_log</code>	Tail live AIStudio backend log — run in a separate tab after <code>ais_start</code>
<code>ais_download_sec_10k</code>	Download SEC 10-K filings from EDGAR to <code>~/Downloads/sec_10k/</code> (~2 GB)
<code>ais_ingest_sec_10k</code>	Ingest SEC 10-K corpus into AIStudio (~30 min, backend must be running)
<code>ais_help</code>	Print this command reference

Every command supports `--help`:

```
ais_start --help
ais_bench --help
```

Upgrading AIStudio

How do I upgrade AIStudio when a new version is released?

Pull the latest code from GitHub:

```
cd ~/Developer/AIStudio && git pull
```

Then update Python dependencies in case new packages were added:

```
source .venv/bin/activate && pip install -r requirements.txt
```

Then restart:

```
ais_start
```

Does upgrading delete my corpus data? No. Your corpus data lives in `~/qdrant_storage/` on disk — completely separate from the AIStudio codebase. It persists across upgrades, restarts, and reboots. You never need to re-ingest after a routine upgrade.

When would I need to re-ingest after an upgrade? Only if the release notes say the chunk format changed. This is rare and always announced. When it happens, run:

```
AI STUDIO VECTORSTORE=qdrant PYTHONPATH=src python3 -m local_llm_bot.app.ingest --corpus <name>
```

The `--force` flag wipes and rebuilds the corpus cleanly.

The demo corpus re-ingests automatically on first `ais_start` after an upgrade if needed.

Corpus Management

How do I ingest a new corpus? Use the UI — open AIStudio, create a new corpus using the **New** button, then upload your files using the **Add** button. AIStudio handles ingestion automatically and shows progress in the chat area.

How do I re-ingest a corpus from scratch? Delete the corpus using the **Delete Corpus** button in the UI (type YES to confirm — it moves to Trash, recoverable). Then create a new corpus with the same name and re-upload your files via **Add**. AIStudio ingests everything in one pass.

What happens when I delete a file from a corpus — is it gone forever? No. Deleted files move to `data/corpora/<name>/uploads/trash/` — not permanently deleted.

How do I recover a file I accidentally deleted from a corpus?

```
# See what's in trash
ls ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/

# Move it back
mv ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/<filename> \
  ~/Developer/AIStudio/data/corpora/<name>/uploads/

# Re-ingest to restore it
```

```
AIStudio_VECTORSTORE=qdrant PYTHONPATH=src python3 -m local_llm_bot.app.ingest \
--corpus <name> --root data/corpora/<name>/uploads --force
```

What happens when I delete an entire corpus — is it gone forever? No. The corpus folder moves to `~/.Trash/`. The folder is renamed to avoid conflicts with previous deletions — it will be named `AIStudio_<name>` or `AIStudio_<name>_<timestamp>` if a folder with that name already exists in Trash.

How do I recover a corpus I accidentally deleted? First find the folder in Trash — it may have a timestamp suffix:

```
ls ~/.Trash/ | grep AIStudio_<name>
```

Then move it back and re-ingest:

```
mv ~/.Trash/AIStudio_<name> ~/Developer/AIStudio/data/corpora/<name>
# Or if it has a timestamp suffix:
mv ~/.Trash/AIStudio_<name>_<timestamp> ~/Developer/AIStudio/data/corpora/<name>
# Then re-ingest
```

Then re-upload the files via the UI (Add button) to restore the corpus in Qdrant.

How do I ingest the SEC 10-K corpus?

The SEC 10-K corpus is a special case. AIStudio provides `ais_download_sec_10k` because the SEC EDGAR filing system uses a specific access protocol — this automates what would otherwise be a complex multi-step download. But the ingest step that follows is **identical to ingesting any corpus you bring yourself**. The download step is what's special.

This is one of the few places in AIStudio where you'll run terminal commands directly. We expose it deliberately — understanding how large corpora are built is part of what makes AIStudio a learning tool, not just a product.

Step 1 — Download the filings to `~/Downloads/sec_10k/` (~5 min, ~2 GB):

```
ais_download_sec_10k
```

Step 2 — Ingest using the AIStudio UI:

Open AIStudio, create a new corpus named `sec_10k`, then upload the files from `~/Downloads/sec_10k/` using the **Upload** button. This is the same process as ingesting any corpus you build yourself — the download step is what's special.

Allow ~34 minutes for ingestion to complete.

Why ~/Downloads? You own the downloaded files — ~2 GB of SEC filings you may want to reuse, inspect, or build a different subset from. They stay in `~/Downloads/sec_10k/` until you decide what to do with them.

Installing and Managing LLMs

How do I see what models are installed?

```
ollama list
```

How do I install a new LLM?

```
ollama pull llama3.1:8b      # ~5GB — recommended default, fast
ollama pull llama3.1:70b     # ~42GB — best quality, requires ~64GB RAM
ollama pull mistral:7b       # ~4.4GB — alternative option
```

Once pulled, the model appears automatically in the **Model** dropdown in the UI. No restart required.

Which model should I choose? On Apple Silicon, `llama3.1:70b` and `llama3.1:8b` have identical query latency (~6–7s) once warm. Choose based on available RAM: - `llama3.1:8b` — 8GB RAM minimum, recommended for most users - `llama3.1:70b` — 64GB+ RAM, best answer quality - `mistral:7b` — good alternative on constrained hardware

How do I remove a model I no longer need?

```
ollama rm mistral:7b
```

See the [Ollama model library](#) for all available models.

Query Settings

The **Query Settings** panel in the sidebar controls how AIStudio retrieves and generates answers. These settings apply to every query until you change them.

Top K — *How many document chunks to retrieve* The number of passages retrieved from your corpus before generating an answer. Higher values give the model more context but increase latency slightly. - Default: `5` — good for most queries - Try `10` for complex questions spanning multiple documents - Try `3` for faster responses on focused questions

Choosing Top K by query type:

Query type	Recommended Top K	Example
Point-in-time, single firm	5	"What was Goldman's CET1 ratio in 2024?"
Trend across multiple years	10–15	"How has JPMorgan's capital ratio changed 2022–2025?"
Cross-firm comparison	10	"Compare risk disclosures at JPMorgan, Goldman, and Citi"
General question	5	"What is the company's AI strategy?"

If an answer seems incomplete or missing years/firms you expected — increase Top K first before rephrasing the question.

Adjust using the **Top K** input field in the sidebar.

Temperature — *How creative vs. precise the answer is* Controls randomness in the model's output. - **0.1–0.3** — precise, factual, stays close to the source documents. Best for financial data, specific numbers, compliance questions. - **0.5–0.7** — more varied, synthesizes across sources. Better for summaries. - Default: **0.3** — recommended starting point Adjust using the **Temperature** input field in the sidebar.

Retrieval Mix — *Balance between literal and conceptual search* Controls how the system finds relevant passages in your corpus.

- **Literal** (slider left, value 1.0) — prioritizes exact word and phrase matching (BM25). Use when you know the specific term, entity name, or abbreviation that appears in your documents. Example: searching for "CET1" by exact term.
- **Conceptual** (slider right, value 0.0) — prioritizes semantic meaning. Finds passages related to your query even when the document uses different phrasing. Example: asking about "capital strength" returns passages about CET1, Tier 1 capital, and capital ratios.
- **Default 0.5** — blends both. Works well for most queries.

Choosing by query type:

Query type	Recommended setting	Why
Named entity, ticker, exact term	Literal (0.7–1.0)	BM25 matches the token directly

Query type	Recommended setting	Why
Thematic or conceptual question	Conceptual (0.0–0.3)	Semantic search handles paraphrase
Multi-firm comparison	Center (0.4–0.6)	Entity names + topic meaning both matter
Single firm, multiple years	Conceptual (0.0–0.2)	Year context is in documents; meaning guides retrieval

Adjust using the **Retrieval Mix** slider in the Query Settings sidebar.

Understanding Citations

Every answer includes numbered citations — [1], [2], [3] — in the text. These refer to the source passages listed in the **References** panel on the right side of the screen.

What citations tell you: - Which document the information came from - Which page in that document

What to do when an answer seems wrong: Check the citations first. If the referenced chunks don't actually contain the claim — the model may be reasoning beyond its sources. Try increasing Top K, adjusting the Retrieval Mix toward Literal, or rephrasing to be more specific.

Uncited claims: if a sentence has no citation marker, the model is drawing on general context rather than a specific retrieved passage. Treat uncited claims with more caution, especially for specific numbers or facts.

Opening the source: click the citation chip in the References panel to see the full chunk text and open the source document.

Benchmark & Corpus Testing

The benchmark harness serves two purposes: - **Performance** — measures query latency per question - **Veracity** — validates answer quality against a questions file

How do I run a benchmark on the demo corpus?

```
ais_bench
```

Parameters (Top K, Temperature, Retrieval Mix, Score Threshold) are read automatically from corpus metadata — no flags needed for built-in corpora.

How do I benchmark a different corpus?

```
ais_bench --corpus sec_10k
ais_bench --corpus help
```

How do I test with a specific model or override settings?

```
ais_bench --corpus demo --model llama3.1:70b
ais_bench --corpus demo --alpha 0.3 --min-score 0.2
ais_bench --help # all flags
```

What makes a benchmark question pass or fail? Three checks — all must pass: 1. All `keywords` appear in the answer (case-insensitive) 2. The answer includes at least one citation 3. The model doesn't hedge with "no information available" or similar phrases

How do I write my own test questions for a corpus? Create `benchmarks/<corpus_name>/<corpus_name>_questions.yaml`:

```
- topic: Your Topic
  questions:
    - id: unique_snake_case_id
      question: The exact question text sent to AIStudio.
      keywords: [term1, term2, term3] # all must appear in the answer to pass
      notes: What a correct answer looks like — which document, what content.
```

Run `ais_bench --corpus <name>` — the questions file is auto-detected by corpus name.

Tips for good keywords: use 2–5 distinctive terms that prove the model retrieved the right content — specific concepts, entity names, or regulatory terms. Avoid generic words that would appear in any answer.

Where do reports go? `benchmarks/<corpus>/reports/` — timestamped `.md`, `.json`, and `.pdf` files. The `.md` shows pass/fail per question with latency, citations, and the answer text.

PDF reports aren't being generated — I see "PDF skipped"

`weasyprint` is required for PDF output and must be installed separately:

```
pip3 install weasyprint --break-system-packages
```

It's already in `requirements.txt` but not installed by default. Once installed, subsequent runs generate `.pdf` reports alongside `.md` and `.json`. The `PDF skipped` message is printed when weasyprint is missing — `.md` and `.json` reports are always generated regardless.

See `TUTORIAL.md §3.4` for a step-by-step walkthrough.

Troubleshooting

UI shows "Error loading corpora" or "Ollama not running" on startup

The browser opened before the backend finished starting. Hard-refresh (`Cmd+Shift+R`). If that doesn't fix it, check the terminal for a Qdrant WAL error (see below).

Ollama version mismatch warning — "client version is X.X.X"

After a Homebrew update, Ollama's CLI binary may update while the background service still runs the old version. You'll see:

```
ollama version is 0.18.0
Warning: client version is 0.22.0
```

Fix:

```
brew services restart ollama
ais_stop
ais_start
```

Qdrant WAL lock — "Can't init WAL: Resource temporarily unavailable"

What it is: A corpus collection's write-ahead log was left locked from an unclean shutdown — force-quit terminal, power loss, or crash mid-write. Qdrant panics on the affected collection at startup. Other collections are unaffected.

How you'll know: Terminal shows the WAL error on `ais_start`, naming the specific collection. UI shows "Error loading corpora." Ingestion immediately fails.

Fix:

```
ais_stop
# Delete only the collection named in the panic message - e.g. aistudio_help
rm -rf ~/qdrant_storage/collections/aistudio_help
ais_start
```

Then re-ingest the corpus via the UI (Add button).

⚠ Delete only the specific collection named in the error — not the entire `~/qdrant_storage/` folder. That would destroy all your corpora.

Prevention: Always stop with `ais_stop`. Never force-quit or close the terminal while AIStudio is running.

Ingestion summary shows wrong file or chunk count

Delete the corpus via the UI and re-ingest. The Qdrant collection may be out of sync with the uploads folder.

How do I rename a corpus? Use the **Rename** button in the corpus header in the UI. AIStudio renames the directory, updates the corpus metadata, and triggers a background re-index. You'll see a progress estimate for large corpora. Do not rename corpus folders manually on disk — use the UI only.

How do I rename a file inside a corpus? AIStudio records file names at ingest time. Renaming a file after ingestion breaks citation lookup — queries will still return results but the "Open ↗" link will fail to resolve.

Safe approach: delete the file from the corpus using the **Delete** button in the Inspect view, then re-upload it with the correct filename via **Add**. AIStudio will re-ingest only the changed file.

For architecture context, see [docs/architecture_decisions.pdf](#). For getting started, see [QUICKSTART.md](#).